

AJOFI

Product Requirements Document (PRD)

Hack4Freedom 2026

1. Executive Summary

Ajofi is a multilingual AI-powered community finance platform designed for informal African workers including market women, traders, artisans, transport workers, and low-income earners.

Ajofi digitizes traditional savings systems such as Ajo, Adashe, and Esusu while preserving the trust structures and community relationships that already exist.

Instead of teaching users Bitcoin first, Ajofi allows users to continue behaving exactly as they already do while Bitcoin, Lightning, Breez SDK, and Nostr operate invisibly in the background.

Users save.

Users contribute.

Users support each other.

Ajofi handles the infrastructure.

Our mission is to make financial coordination, community savings, and emergency support accessible to millions of Africans who remain excluded from formal financial systems.

2. Problem Statement

Across Africa, millions of informal workers save money every day.

Market women.

Food vendors.

Tailors.

Hairdressers.

Artisans.

Transport workers.

These individuals already possess financial discipline.

The problem is not that they do not save.

The problem is that existing financial systems were never designed around their behavior.

Traditional savings circles face several challenges:

- Missing records
- Collector fraud
- Poor transparency
- Language barriers
- Financial exclusion
- Lack of emergency support
- Dependence on cash
- Limited access to digital tools

Most fintech products target salaried workers.

Ajofi targets the informal economy.

3. User Research & Market Validation

To understand the problem deeply, our team conducted direct interviews with market women and traders.

The objective was to understand:

- How people save
- Why they trust traditional Ajo systems
- What frustrations exist
- What would make them adopt digital alternatives

Key Findings

Finding 1

Daily and monthly savings systems are already deeply embedded in community culture.

Interview Response:

"We make monthly contributions and daily contributions."

Finding 2

Ajo helps enforce financial discipline.

Interview Response:

"It helps a person save money properly."

Finding 3

Trust is more important than technology.

Interview Response:

"She is reliable."

Finding 4

Many participants intentionally choose later collection slots because it helps them accumulate larger amounts.

Interview Response:

"When you finally collect it, it feels bigger and you can use it for what you really need."

Finding 5

Community-based savings provide peace of mind.

Interview Response:

"Ajo in the market brings peace and helps people."

4. Survey Insights

The interviews revealed five major user needs:

Transparency

Users want visibility into contributions and balances.

Trust

Users want proof that contributions are safe.

Accessibility

Users want systems that work with their language and literacy level.

Simplicity

Users do not want complicated financial applications.

Emergency Support

Users want help during unexpected financial crises.

5. Target Users

Primary Users

Market Women

Daily cash-based traders.

Food Vendors

Small-scale sellers operating in local communities.

Artisans

Tailors, mechanics, welders, hairdressers, carpenters.

Transport Workers

Drivers and informal transportation operators.

6. User Personas

Persona 1 — Amina

Age: 42

Location: Kano

Occupation: Tomato Seller

Characteristics:

- Participates in Adashe
- Uses WhatsApp
- Limited English literacy
- Earns daily income

Goals:

- Save consistently
- Avoid fraud
- Access emergency support

Persona 2 — Funke

Age: 38

Location: Lagos

Occupation: Market Trader

Goals:

- Track contributions
- Improve transparency
- Build trust

7. Product Vision

To become the financial operating system for informal African communities.

Ajofi transforms trust, savings, and community coordination into digital infrastructure while preserving cultural financial practices.

8. Solution Overview

Ajofi combines:

- Multilingual AI
- Savings Circles
- Emergency Support Pools
- Bitcoin Infrastructure
- Nostr Trust Layer
- USSD Access

into one unified platform.

9. Core Features

Phone Number Registration

Users register using:

- Phone Number
- OTP Verification

No wallet setup required.

Multilingual AI Assistant

Supported Languages:

- English
- Yoruba
- Hausa
- Igbo
- Pidgin

Functions:

- User onboarding
- Product guidance
- Financial education
- Voice interaction
- Navigation support

The AI behaves like a trusted community assistant.

Savings Circles

Users can:

- Create circles
- Join circles
- Track contributions
- Manage collection schedules

Emergency Support Pools

Members contribute to emergency funds.

Funds can be requested for:

- Health emergencies

- School fees
- Business recovery

Trust Scores

Trust scores are generated from:

- Contribution history
- Circle participation
- Completion records

USSD Access

Users without smartphones can access:

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1. Save Money
2. Check Balance
3. My Circle
4. Emergency Support
5. Change Language

10. User Flow

Registration Flow

Phone Number



OTP Verification



Profile Creation



Language Selection



Dashboard

Circle Creation Flow

Create Circle



Set Contribution Amount



Invite Members



Members Join



Circle Activated

Contribution Flow

Deposit Naira



Payment Verification



Contribution Confirmed



Circle Updated



Trust Record Stored

Collection Flow

Collection Day Trigger



Beneficiary Identified



Funds Released



Collection Recorded



Members Notified

11. AI Architecture

Ajofi's AI layer serves as the primary accessibility engine.

Components:

Translation Layer

Real-time translation between:

- English
- Yoruba
- Hausa
- Igbo
- Pidgin

Conversational Layer

Natural language interaction.

Example:

"Save ₦2,000 today."

"Show my circle balance."

"When is my collection day?"

Financial Education Layer

Explains concepts using local language and cultural context.

The goal is to educate users through usage rather than tutorials.

12. Bitcoin Infrastructure

Bitcoin operates as invisible infrastructure.

Users never interact with:

- Wallet addresses
- Sats
- Lightning invoices
- Seed phrases

Users only see Naira.

Bitcoin provides:

- Settlement
- Security
- Portability
- Scalability

13. Breez SDK Architecture

Purpose:

Provide Lightning infrastructure without operating our own node.

Contribution Flow:

User Deposit

↓

Paystack

↓

Bitnob Conversion

↓

Breez Treasury Wallet

↓

Circle Pool

Collection Flow:

Circle Pool

↓

Breez Lightning Payment

↓

Bitnob Off-Ramp

↓

Bank Account

Benefits:

- Instant settlement
- Lower costs
- Cross-border compatibility
- Bitcoin-native infrastructure

14. Nostr Architecture

Purpose:

Provide decentralized trust infrastructure.

Components:

Identity

Portable user identity.

Contribution Records

Signed immutable records.

Circle Groups

Community coordination.

Reputation

Trust history portable across circles.

NIPs Used:

- NIP-29
- NIP-58
- NIP-78

15. System Architecture

Frontend

React Native

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Backend API

Node.js + TypeScript

↓

Services Layer

Paystack

Bitnob

Breez SDK

Nostr

↓

Database

PostgreSQL

↓

Caching

Redis

16. Database Design

Users

- id
- phone
- name
- language
- trust_score

Circles

- id
- name
- contribution_amount
- member_limit

CircleMembers

- id
- user_id
- circle_id

Contributions

- id
- amount
- status
- created_at

Collections

- id
- circle_id
- beneficiary_id
- amount

EmergencyRequests

- id
 - user_id
 - amount
 - reason
-

17. API Documentation

Authentication

POST /auth/send-otp

Request

```
{  
  "phone": "+234xxxxxxxxxx"  
}
```

POST /auth/verify-otp

Request

```
{  
  "phone": "+234xxxxxxxxxx",  
  "code": "123456"  
}
```

POST /auth/profile

Creates user profile.

POST /auth/bank

Links bank account.

Circles

GET /circles

Returns user circles.

POST /circles

Creates circle.

GET /circles/:id

Returns circle dashboard.

GET /circles/:id/members

Returns members.

POST /circles/join/:inviteCode

Join circle.

Contributions

GET /contributions/:circleId/instructions

Returns payment instructions.

Collections

POST /collections/:circleId/trigger

Triggers collection payout.

Savings

GET /users/me/savings

Returns savings summary.

18. Security

Authentication:

- OTP Verification
- JWT Sessions

Data Protection:

- Encryption at Rest
- Secure API Access

Trust Protection:

- Nostr Signed Events

Payments:

- Breez Lightning Infrastructure

19. Business Model

Revenue Sources

Transaction Fees

Small percentage of contributions.

Premium Circle Features

Advanced reporting and management.

Cooperative Partnerships

Women's groups.

Trade associations.

Community organizations.

Embedded Financial Services

Insurance.

Credit.

Business support tools.

20. SDG Alignment

SDG 1 — No Poverty

SDG 5 — Gender Equality

SDG 8 — Decent Work and Economic Growth

SDG 9 — Industry Innovation and Infrastructure

SDG 10 — Reduced Inequalities

21. Roadmap

Phase 1

Savings Circles

Phase 2

Emergency Pools

Phase 3

Multilingual AI Expansion

Phase 4

USSD Rollout

Phase 5

Portable Reputation System

Phase 6

Cross-Border Community Finance

22. Conclusion

Ajofi is not attempting to replace traditional community finance.

Ajofi strengthens what already works.

By combining multilingual AI, community savings behavior, Bitcoin infrastructure, Breez SDK, and Nostr-based trust systems, Ajofi enables millions of informal workers to participate in a safer, more transparent, and more inclusive financial future.